

Does regional variation in wage levels identify the effects of a national minimum wage?

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Insper

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STATISTICAL MODELS AND SHOE LEATHER

*David A. Freedman**

Regression models have been used in the social sciences at least since 1899, when Yule published a paper on the causes of pauperism. Regression models are now used to make causal arguments in a wide variety of applications, and it is perhaps time to evaluate the results. No definitive answers can be given, but this paper takes a rather negative view. Snow's work on cholera is presented as a success story for scientific reasoning based on nonexperimental data. Failure stories are also discussed, and comparisons may provide some insight. In particular, this paper suggests that statistical technique can seldom be an adequate substitute for good design, relevant data, and testing predictions against reality in a variety of settings.

Causal Claims

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 **Causal Claims**

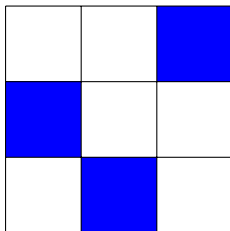
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WHAT IS A “CLAIM GRAPH”?

- A claim graph is a paper-level, machine-readable map of what a paper states.
- Nodes are standardized economic concepts (mapped to JEL codes). Directed edges are the relationships the authors claim between concepts.
- Each edge is annotated with its evidentiary basis, including whether it is supported by credible causal designs (e.g., DiD, IV, RDD, RCTs, event studies, synthetic control) or by non-causal evidence.

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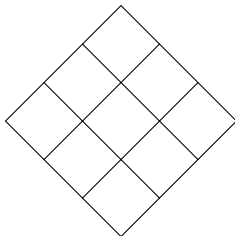
Identifying minimum wage effects



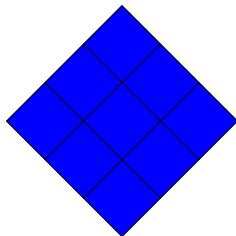
Minimum wage increases in blue states

$$y_{r,t} = \alpha_r + \delta_t + \beta T_r \cdot \mathbf{1}\{t \geq 1\} + \epsilon_{r,t}$$

A national change in the minimum wage?

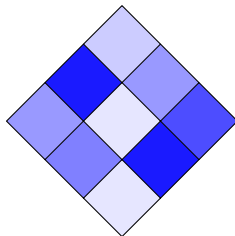


A national change in the minimum wage?



$$y_{r,t} = \alpha_r + \delta_t + \beta T_{r,t} \cdot \mathbf{1}\{t \geq 1\} + \epsilon_{r,t}$$

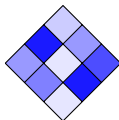
A national change in the minimum wage?



$$y_{r,t} = \alpha_r + \delta_t + h_t(E_{r,t}) + \epsilon_{r,t}$$

- Exposure $E_{r,t}$ is constructed from observed wage distributions

A national change in the minimum wage?



- Two main approaches:
 - Effective minimum wage designs (Lee 1999)
 - Fraction Affected/Gap designs (Card 1992)
- Still actively used in contexts without regional variation
 - Engbom and Moser (2022), Dustmann et al. (2022), Giupponi et al. (2024)...

This paper

Does regional variation in wage levels identify the effects of a national minimum wage?

- Uses theory to show exposure-based regressions are often misspecified
- Quantifies biases using simulations
- Discusses these econometric problems in the context of Brazil
- Distills the evidence into practical recommendations

Bottom line: avoid the Effective-Minimum-Wage design; use Fraction Affected/Gap designs only when their assumptions can be tested and defended.

Outline

- ➊ **Setup and definitions**
- ➋ Effective minimum wage design
- ➌ Fraction affected/gap designs
- ➍ Extended simulations
- ➎ Brazil
- ➏ Recommendations for applied researchers
- ➐ Broader implications

The data-generating process

$$\mathbf{y}_{r,t} = f(mw_t, \boldsymbol{\theta}_{r,t}) \quad [\boldsymbol{\theta}'_{r,0}, \boldsymbol{\theta}'_{r,1}]' \sim G$$

- $r \in \{1, \dots, R\}$ indexes regions, and $t \in \{0, 1\}$ indexes time
- Outcomes $\mathbf{y}_{r,t}$: employment-to-population ratio and log-wage quantiles
- f is an economic model
- $mw_1 > mw_0$
- $\boldsymbol{\theta}_{r,t}$: region-time-specific parameters—including the error term in the regression

The data-generating process

$$\mathbf{y}_{r,t} = f(mw_t, \boldsymbol{\theta}_{r,t}) \quad [\boldsymbol{\theta}'_{r,0}, \boldsymbol{\theta}'_{r,1}]' \sim G$$

$$\mathbf{ATE}_0 = E_G[f(mw_1, \boldsymbol{\theta}_{r,0})] - E_G[\mathbf{y}_{r,0}]$$

$$\mathbf{ATE}_1 = E_G[\mathbf{y}_{r,1}] - E_G[f(mw_0, \boldsymbol{\theta}_{r,1})]$$

$$\mathbf{ATE} = \frac{\mathbf{ATE}_0 + \mathbf{ATE}_1}{2}$$

Expectation with respect to the distribution G

The data-generating process

$$\mathbf{y}_{r,t} = f(mw_t, \boldsymbol{\theta}_{r,t}) \quad [\boldsymbol{\theta}'_{r,0}, \boldsymbol{\theta}'_{r,1}]' \sim G$$

What this rules out:

- Designs at the firm or worker level: Harasztosi and Lindner (2019), Dustmann et al. (2022)
- Regional spillovers/migration: Cadena (2014), Huang (2019)
- Short vs. long effects of minimum wages: Sorkin (2015), Meer and West (2016), Vogel (2023)
- Measurement error in region-level statistics: Autor, Manning, and Smith (2016)
- Staggered d-in-d: de Chaisemartin and D'Haultfoeuille (2020)

Exposure-based regressions

$$y_{r,t} = \alpha_r + \delta_t + h_t(E_{r,t}) + \epsilon_{r,t}$$

Potential advantages:

- 1 Researchers can be agnostic about theoretical mechanisms.
- 2 These regressions are readily accepted as “reduced-form” or “data-driven” approaches.

Potential problems:

- 1 The exposure variable may not be observable.
- 2 Exposure may not be one-dimensional.
- 3 The response function $h_t(\cdot)$ may be misspecified.

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Effective minimum wage design

Ordinary least squares with the following specification:

$$y_{r,t} = \alpha_r + \delta_t + \beta [mw_t - w_{0.5,r,t}] \\ + \gamma [mw_t - w_{0.5,r,t}]^2 + \epsilon_{r,t}$$

- Outcomes are often wage inequality measures, e.g.,
 $w_{0.10,r,t} - w_{0.5,r,t}$
 - $w_{q,r,t}$: quantile q of the log-wage distribution
- Also used for evaluating employment effects
- $[mw_t - w_{0.5,r,t}]$ is the *effective minimum wage*
- Average effects are calculated based on $\hat{\beta}$ and $\hat{\gamma}$

Lee's (1999) semiparametric model

The CDF of *latent* log wages has the form:

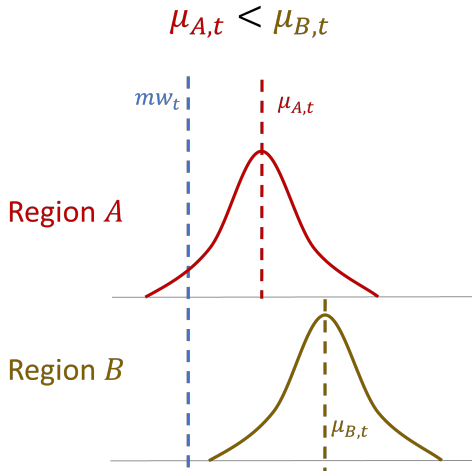
$$F_t \left(\frac{w - \mu_{r,t}}{\sigma_{r,t}} \right) \quad \text{with } F_t(0) = 0.5$$

- $\mu_{r,t}$ is the location (or *centrality*) parameter
- $\sigma_{r,t}$ is the *dispersion* parameter

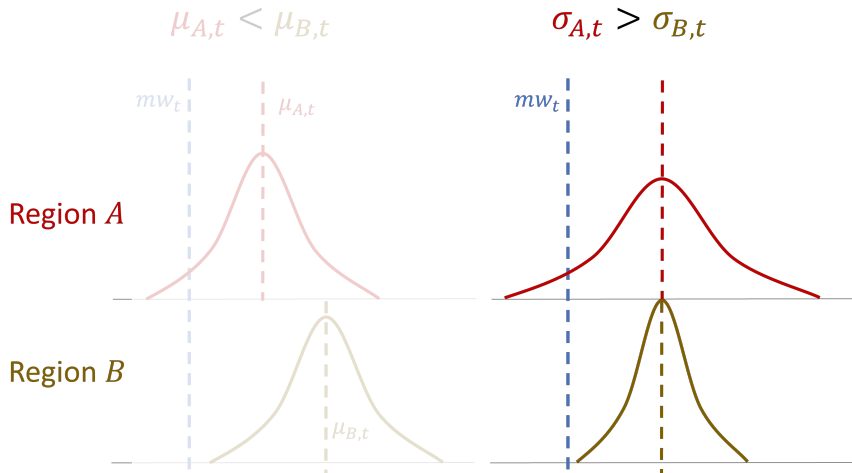
Lee's key assumption: $\mu_{r,t}$ and $\sigma_{r,t}$ are uncorrelated.

If this assumption holds, where does identification come from?

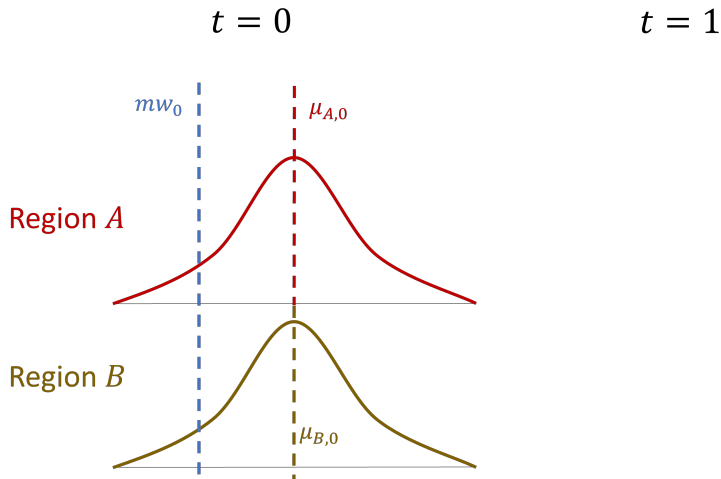
Location (centrality) and dispersion



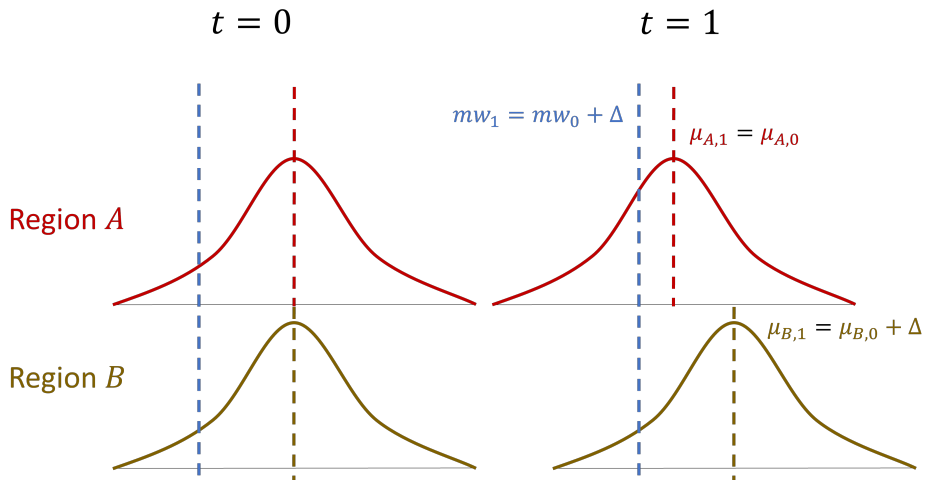
Location (centrality) and dispersion



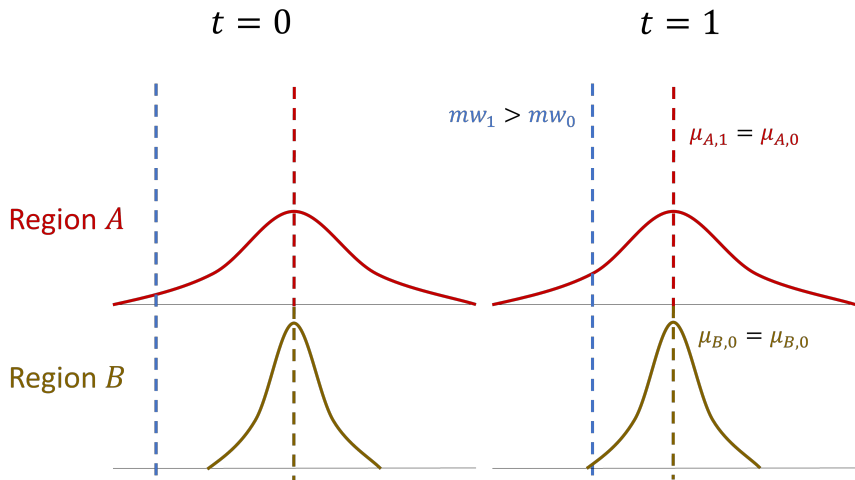
The ideal scenario: A is treated, B is not



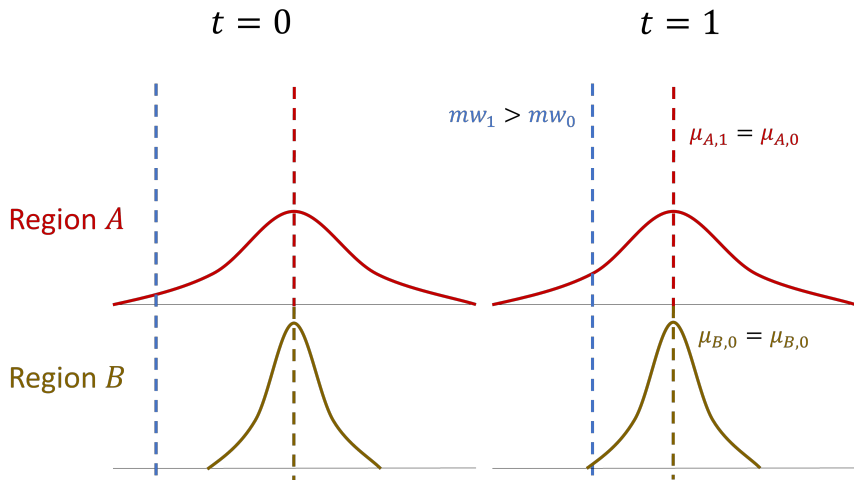
The ideal scenario: A is treated, B is not



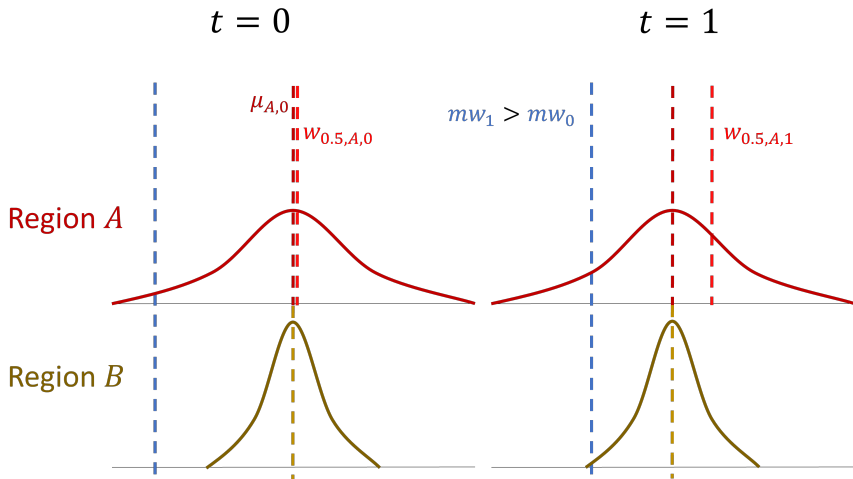
Alternative scenario: no idiosyncratic shocks



If we could observe $\mu_{r,t}$: no variation



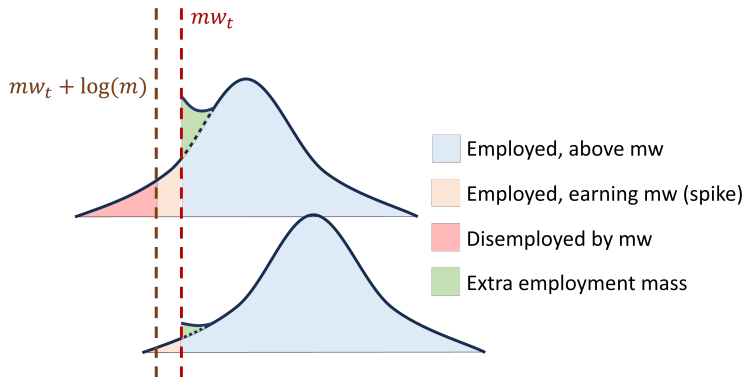
Min. wage effects on the median: bad variation



Correlated measurement errors

- Treatment intensity should be $mw_t - \mu_{r,t}$, but instead we use $mw_t - w_{0.5,r,t}$
- The “measurement error” $w_{0.5,r,t} - \mu_{r,t}$ is **not** classical; rather, it correlates with the outcomes of interest
- A shock to $\sigma_{r,t}$ also creates problems (see paper)
- Next step: simulations

Simulations: the Normal-markdown model



- Markdown parameter $m = 0.7$ in the baseline calibration
- Log minimum wage increases by 0.2 unless otherwise noted.

Log min. wage increases by 0.2

Averages across 1,000 simulations, 200 regions in each simulation.

Panel A has $\sigma_{r,t} = \bar{\sigma}_t$ and $\text{Corr}(\mu_{r,0}, \mu_{r,1}) = 0.89$

	Outcome			
	Emp.	p10 - p50	p25 - p50	p90 - p50
<i>Panel A: Regions differ only in location</i>				
True average causal effect	-0.010	0.019	0.006	-0.004
Effective min. wage	-0.010	0.020	0.006	-0.004
	(0.000)	(0.001)	(0.000)	(0.000)

Log min. wage increases by 0.2

Averages across 1,000 simulations, 200 regions in each simulation.

$\text{Corr}(\mu_{r,t}, \sigma_{r,t}) = 0$, satisfying Lee's assumptions

	Outcome			
	Emp.	p10 - p50	p25 - p50	p90 - p50
<i>Panel B: Regions differ in location and dispersion</i>				
True average causal effect	-0.010	0.020	0.006	-0.004
Effective min. wage	-0.007	0.034	0.015	-0.023
	(0.002)	(0.011)	(0.007)	(0.014)

Larger minimum wage hike (0.4 vs. 0.2)

Averages across 1,000 simulations, 200 regions in each simulation.

	Outcome			
	Emp.	p10 - p50	p25 - p50	p90 - p50
<i>Panel C: As above, but larger increase in min. wage</i>				
True average causal effect	-0.032	0.078	0.017	-0.012
Effective min. wage	-0.014	0.117	0.046	-0.080
	(0.007)	(0.020)	(0.012)	(0.026)

Issue #2: Correlation between $\mu_{r,t}$ and $\sigma_{r,t}$

	Outcome			
	Emp.	p10 - p50	p25 - p50	p90 - p50
<i>Panel B: Contemporaneous correlation of 0.076</i>				
True average causal effect	-0.010	0.020	0.006	-0.004
Effective min. wage	-0.002	0.077	0.040	-0.076
	(0.002)	(0.010)	(0.006)	(0.013)

Correlations between the mean and standard deviation of log wages across regions:

- US states: 0.076 in 1989, 0.264 in 2004
- Brazilian microregions: 0.339; 0.528 where the minimum wage barely binds

Other specifications and diagnostics

- ❶ Are there biases if average employment effects are zero or positive? **Yes** (will show later).
- ❷ Adding trends or removing time fixed effects? **No.** [▶ Table](#)
- ❸ Using higher quantiles as the deflator? **No.** [▶ Table](#)
- ❹ IV strategy by Autor, Manning, and Smith (2016)? [▶ Table](#)
 - **Yes**, but only feasible with state-level minimum wages.
 - The version with two rather than three instruments may be better.
- ❺ Diagnosing with upper-tail spillovers?
 - Subject to false positives and false negatives. [▶ Why?](#)

Taking stock: effective minimum wage

- Identification requires latent location shocks that move wages but not employment or the shape of the wage distribution.
- Even when Lee's location-dispersion orthogonality holds and $\text{Corr}(w_{50}, \mu) > 0.99$, correlated proxy error creates meaningful bias.
- Small, empirically realistic location-dispersion correlations can make the bias much larger.
- Higher-quantile deflators, region-specific trends, and upper-tail diagnostics do not reliably solve the problem.

Recommendation: do not use this design.

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Fraction affected and Gap designs

$$y_{r,t} = \alpha_r + \delta_t + \beta E_r \mathbf{1}\{t = 1\} + \epsilon_{r,t}$$

The time-invariant exposure measure E_r is calculated using pre-reform data.

Fraction affected

$$FA_r = \frac{1}{l_r} \sum_i \mathbf{1}\{w_{i,0} < mw_1\}$$

Gap

$$Gap_r = \frac{\sum_i \max\{e^{mw_1} - e^{w_{i,0}}, 0\}}{\sum_i e^{w_{i,0}}}$$

Issue #1: nonlinear exposure-response relationships

- Papers using fraction-affected/gap designs assume linear effects...
- But minimum wage effects are never linear in economic models
 - Employment effects may initially be negligible, then become positive, then negative
 - Wage compression is typically positive but varies in intensity
- Test: Do linear regressions recover the true causal effects in a **perfect** application?

Misspecification in ideal applications

- Regions differ only in a time-invariant location parameter; the parallel-trends assumption holds by construction.
- The model includes only truncation and censoring effects

	Emp.	p10	p25	p50	p90
<i>Panel A: Small initial minimum wage; truncation/censoring only</i>					
True effect	-0.006	0.016	0.008	0.004	0.002
Fraction affected	-0.008	0.020	0.010	0.006	0.003
Gap	-0.006	0.015	0.007	0.004	0.002
<i>Panel B: Large initial minimum wage; truncation/censoring only</i>					
True effect	-0.031	0.117	0.036	0.020	0.010
Fraction affected	-0.039	0.185	0.044	0.026	0.013
Gap	-0.028	0.127	0.031	0.019	0.009

Misspecification in ideal applications

- The model now allows for positive employment effects

	Emp.	p10	p25	p50	p90
<i>Panel C: Small initial minimum wage; added employment mass</i>					
True effect	0.010	-0.003	-0.012	-0.006	-0.003
Fraction affected	0.002	0.067	-0.002	-0.001	-0.000
Gap	0.001	0.052	-0.001	-0.001	-0.000
<i>Panel D: Large initial minimum wage; added employment mass</i>					
True effect	-0.002	0.148	0.038	0.002	0.001
Fraction affected	-0.038	0.132	0.143	0.026	0.012
Gap	-0.027	0.091	0.101	0.019	0.008

- Note: biases need not have the same sign.

Nonlinearity and misspecification: discussion

- We will test this issue using a broader range of DGPs later in the presentation
- Solutions discussed in the literature:
 - Allowing for heterogeneity by groups such as industries (Dube and Lindner 2024)
 - “Bin-level” designs (e.g., Giupponi 2024)
 - Neither approach solves the underlying conceptual problem.
- Solution proposed here: use polynomial specifications
 - Polynomial specifications are not guaranteed to help, but they can reduce biases in several DGPs tested in the paper.

Issue #2: Deviations from parallel trends

- Regression to the mean (especially relevant with finely disaggregated regions or groups).
- Regional convergence.
- **Newly described source:** trends in the dispersion of latent log wages (potentially arising from technical change).
- Adding region-specific time trends can make the pre-trends **look** better but can magnify biases

Issue #2: Deviations from parallel trends

Good news:

- These issues may be detected using tests for parallel pre-trends if they are smooth over time
- To detect regression to the mean, the exposure measure must be constructed using a single period (papers sometimes average exposure across all pre-reform years).

Bad news:

- Parallel pre-trends cannot be tested without a pre-reform period with stable minimum-wage levels.

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How common are those biases?

- 500 DGPs in the Normal-markdown class; 1,000 samples of 200 regions per DGP
 - Positive, negative, or small employment effects; varying wage spillovers
 - Parameters centered on US regional data; location-dispersion correlation capped at 0.20
- I also analyze a “canonical” model with endogenous returns to skill (Katz and Murphy 1992)
- Main takeaway: economically and statistically significant biases are the norm rather than the exception.

Which FA/Gap specification performs best?

500 additional DGPs for which the parallel-trends assumption holds by construction:

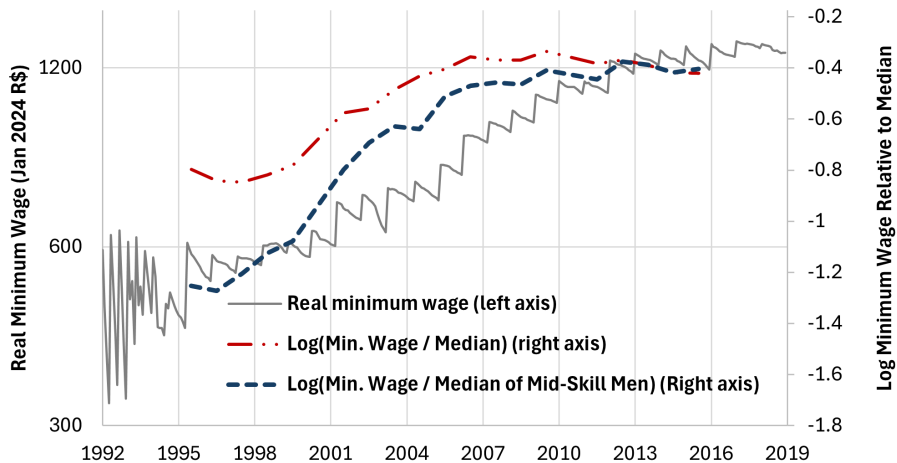
Mean absolute bias	Emp.	p10-p50	p25-p50	p90-p50
Fraction affected, linear	0.013	0.104	0.055	0.012
Gap, linear	0.008	0.087	0.027	0.007
Fraction affected, cubic	0.003	0.183	0.104	0.017
Gap, quadratic	0.006	0.081	0.033	0.004

- Employment: cubic polynomial in fraction affected
- Wage outcomes: quadratic polynomial in gap
- One specification for all outcomes: quadratic gap

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The rise of the Brazilian minimum wage



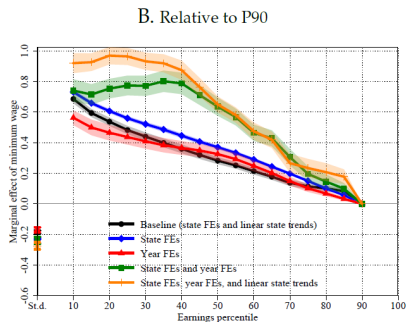
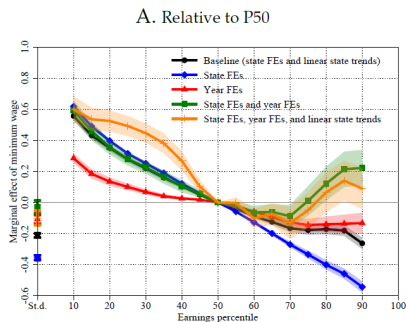
Why Brazil is an informative application

- ① A large country with substantial variation across local labor markets
- ② Essentially no usable regional variation in minimum-wage laws
- ③ Several papers use regional wage exposure to study the same national increase

First step: summarize results from existing papers exploring the post-Plano Real period.

Paper	Data	Preferred Specification	Results
Hinojosa (2019)	PME, PNAD	Eff. min. wage (p60), region and time FEs, region-specific trends.	Wage compression for all workers; spillovers up to the median.
Saltiel and Urzua (2022)	RAIS	Eff. min. wage (p50); region and time FEs; region-specific trends.	No effects on employment.
Engbom and Moser (2022)	RAIS	Eff. min. wage based on p90, region FEs, region-specific trends.	Wage compression for formal workers; spillovers up to p80.
	PNAD	As above, but based on p50.	No employment effects.
	PME	As above, but based on p50.	No effects on transitions between formal and informal sectors.
Neumark, Cunningham and Siga (2006)	PME	“Rolling” fraction-affected, region and time FEs, includes lags.	Large reductions in employment and household income for families in the bottom 30%.
Derenoncourt et al. (2025)	PNAD	Fraction-affected at the state×industry×formality level; state-industry and industry-year FEs.	Wage increases for formal and informal workers; modest reallocation toward the informal sector.
Parente, Brotherhood and Iachan (2025)	PNAD	Binary version of fraction-affected.	Large increases in informality and wage dispersion.

Within-paper robustness



Copy of Figure B.14 in Engbom and Moser (2022).

Regressions using 151 microregions, 1998 and 2012

The outcome is **formal** employment as a share of the working-age population.

	Emp.
Structural model	-0.033

Regressions using 151 microregions, 1998 and 2012

The outcome is **formal** employment as a share of the working-age population.

	Emp.
Structural model	-0.033
Fraction affected	-0.019 (0.007)
Binary measure, 50% treated	-0.004 (0.004)
Effective min. wage	-0.061 (0.022)
Effective min. wage, no region FE	-0.090 (0.009)
Effective min. wage, no time FE	0.101 (0.004)

Brazil: takeaways

- The existing literature contains incompatible results, even though the studies presumably use the same source of identifying variation
- Impartial observers have no clear way to tell which specification is more credible
- **We do not have a minimum-wage quasi-experiment in Brazil**
 - A gradual increase rather than a sharp reform
 - A multitude of confounding trends
- Can we control for other factors, such as education?
 - Yes—but then we are doing structural modeling.

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Recommendations for applied researchers

What should researchers do when there is no regional variation in minimum wages?

- 1 Do not use the effective-minimum-wage design.
- 2 Do not use fraction-affected/gap designs unless you can test for parallel pre-trends.
- 3 Define the treatment variable using a single pre-treatment year.
- 4 Use a few institutionally motivated baseline controls interacted with flexible time effects; do not use unit-specific trends.
- 5 Use cubic fraction affected for employment and quadratic gap for wages; if one specification is required, use quadratic gap.

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Broader relevance

The core issue is not unique to minimum wages: an observed local equilibrium statistic is often treated as if it were a primitive shock.

- **Local exposure to recessions:** realized employment losses combine the aggregate downturn with local adjustment margins; predetermined exposure still imposes a response function.
- **Employer concentration:** a local HHI reflects entry, exit, mergers, demand, productivity, mobility, and market boundaries; identical changes can represent different shocks.

Give exposure measures economic content, make the identifying variation explicit, and test the implied restrictions.

OLS estimates for the US, 1972-2012

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Panel C. Males and females—state fixed effects and trends

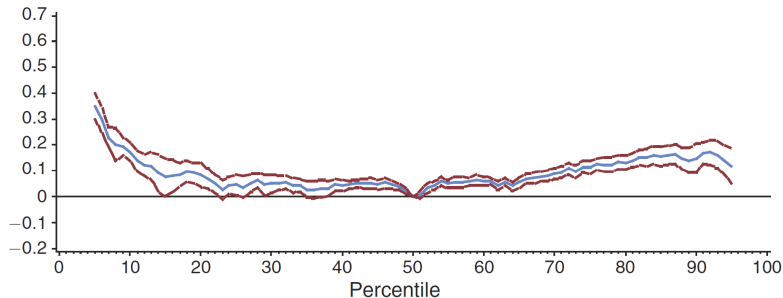


FIGURE 3. OLS ESTIMATES OF THE RELATIONSHIP BETWEEN $\log(p) - \log(p50)$ AND $\log(\min) - \log(p50)$ AND ITS SQUARE, 1979–2012

- From Autor, Manning and Smith (2016)

Fix with regional trends/controls?

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- Options include controlling for worker composition and regional trends.
- This only helps if *residual variation* in $\mu_{r,t}$ is uncorrelated with $\sigma_{r,t}$.

Lee (1999) argues *fewer* controls can be better:

"... the reduced identifying variation resulting from eliminating the "permanent" state effects may magnify biases due to misspecification, in the same way biases stemming from measurement error in the independent variable are magnified when true variation in the independent variable is reduced."

- In simulations, the estimator without region fixed effects works well for wage spillover effects, but not for employment.

Removing time fixed effects

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	Outcome			
	Emp.	p10 - p50	p25 - p50	p90 - p50
True average causal effect	-0.010	0.020	0.006	-0.004
Effective min. wage	-0.007	0.034	0.015	-0.023
	(0.002)	(0.011)	(0.007)	(0.014)
Effective min. wage, no region FE	-0.010	0.022	0.007	-0.007
	(0.001)	(0.004)	(0.002)	(0.005)
Effective min. wage, no time FE	-0.007	0.052	0.024	-0.041
	(0.001)	(0.003)	(0.002)	(0.004)

Percentile 90 as the deflator [◀ Back](#)

	Outcome			
	Emp.	p10 - p90	p25 - p90	p50 - p90
True average causal effect	-0.010	0.024	0.009	0.004
Effective min. wage, p90	0.009	0.220	0.177	0.119
	(0.002)	(0.013)	(0.011)	(0.007)

State-level variation and IV approaches [◀ Back](#)

		Outcome		
	Emp.	p10 - p50	p25 - p50	p90 - p50
<i>Panel A: No regional variation in minimum wage.</i>				
True average causal effect	-0.010	0.020	0.006	-0.004
Effective min. wage	-0.002	0.076	0.040	-0.075
	(0.002)	(0.010)	(0.006)	(0.013)
<i>Panel B: 20% of regions with local min. wage</i>				
True average causal effect	-0.015	0.035	0.008	-0.005
Effective min. wage	-0.015	0.050	0.015	-0.018
	(0.001)	(0.005)	(0.003)	(0.006)
Two instruments	-0.015	0.036	0.009	-0.006
	(0.002)	(0.006)	(0.004)	(0.008)
Three instruments (AMS)	-0.017	0.041	0.008	-0.005
	(0.002)	(0.005)	(0.003)	(0.006)
<i>Panel C: 40% of regions with local min. wage</i>				
True average causal effect	-0.020	0.053	0.011	-0.007
Effective min. wage	-0.019	0.059	0.015	-0.016
	(0.001)	(0.004)	(0.002)	(0.005)
Two instruments	-0.020	0.051	0.011	-0.007
	(0.002)	(0.004)	(0.002)	(0.006)
Three instruments (AMS)	-0.021	0.053	0.010	-0.007
	(0.001)	(0.004)	(0.002)	(0.005)

Subject to false positives:

- Models such as Engbom and Moser (2022) and Haanwinckel (2025) have mechanisms that generate spillovers high into the wage distribution, such as reallocation to higher-wage firms (as in Dustmann et al. 2022).

Subject to false negatives:

- The estimator may be biased for employment and lower-tail spillovers but unbiased for upper-tail spillovers, especially when the effects of the issues discussed here are combined with measurement-error bias from Autor, Manning, and Smith (2016).

		Outcome			
	Emp.	p10 - p50	p25 - p50	p90 - p50	
<i>Panel A: Initial minimum wage is low, elasticity = 3.0</i>					
True average causal effect	-0.009	0.017	0.005	-0.004	
Eff. min. wage, no region FE	-0.009	0.018	0.003	0.044	
	(0.000)	(0.001)	(0.001)	(0.002)	
<i>Panel B: Initial minimum wage is low, elasticity = 1.4</i>					
True average causal effect	-0.009	0.016	0.005	-0.003	
Eff. min. wage, no region FE	-0.010	0.002	-0.007	0.078	
	(0.000)	(0.002)	(0.001)	(0.002)	
<i>Panel D: Initial minimum wage is high, elasticity = 1.4</i>					
True average causal effect	-0.039	0.054	0.019	-0.016	
Eff. min. wage, no region FE	-0.045	0.042	0.008	0.066	
	(0.001)	(0.001)	(0.001)	(0.002)	

Brazilian structural model comparison

[◀ Back](#)

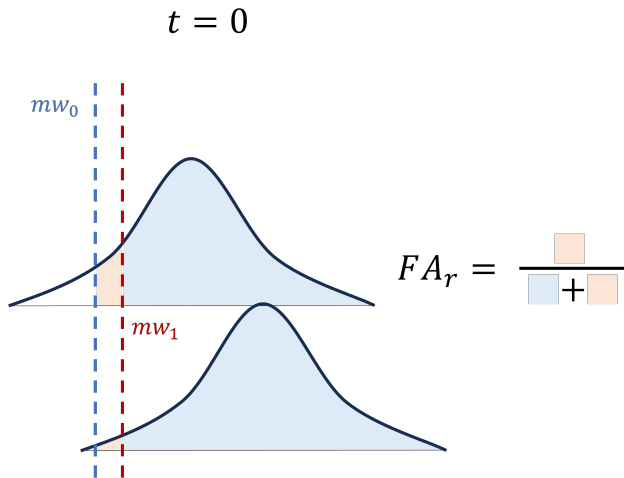
	Emp.	Outcome		
		p10 - p50	p25 - p50	p90 - p50
Structural model	-0.033	0.208	0.070	0.005
Effective min. wage	-0.061	0.188	0.127	0.076
	(0.022)	(0.020)	(0.015)	(0.033)
Effective min. wage, no region FE	-0.090	0.158	0.091	-0.059
	(0.009)	(0.014)	(0.009)	(0.022)
Effective min. wage, no time FE	0.101	0.200	0.103	-0.127
	(0.004)	(0.006)	(0.005)	(0.013)
AMS, no time FE	0.118	0.201	0.101	-0.149
	(0.006)	(0.006)	(0.005)	(0.013)

Structural estimates are from Haanwinckel (2025).

$$y_{o,r,t} = \alpha_{o,r} + \delta_{o,t} + \beta_o FA_r \cdot \mathbf{1}\{t = 1\} + \epsilon_{o,r,t}$$

- Each outcome o is analyzed in a separate regression
- FA_r : share of workers i with $mw_0 \leq w_i < mw_1$ at time $t = 0$
- $\hat{\mathbf{ATE}}_o$ is the product of the average value of FA and β_o
- Typical application: a one-time mw hike following years of stability
 - This setting allows tests for differential pre-trends

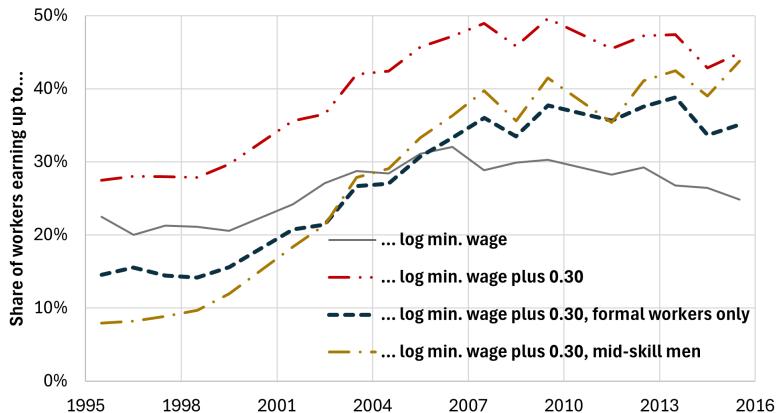
Fraction affected design [◀ Back](#)



$$y_{o,r,t} = \alpha_{o,r} + \delta_{o,t} + \beta_o \text{Gap}_r \cdot \mathbf{1}\{t = 1\} + \epsilon_{o,r,t}$$
$$\text{Gap}_r = \frac{\sum_{i=1}^{I_r} \max\{\exp(mw_1) - \exp(w_{i,0}), 0\}}{\sum_{i=1}^{I_r} \exp(w_{i,0})}$$

- Introduced by Card (1994) at the firm level; later used at the regional level (e.g., Dustmann et al., 2022)

Brazil: alternative bindingness measures



Brazil estimates: fraction affected

	Emp.	Outcome			
		p10	p25	p50	p90
Structural model	-0.033	0.297	0.159	0.090	0.095
Fraction affected	-0.019	0.248	0.235	0.239	0.183
	(0.007)	(0.013)	(0.014)	(0.015)	(0.020)
Quadratic on FA	-0.012	0.350	0.304	0.267	0.184
	(0.013)	(0.016)	(0.021)	(0.028)	(0.037)
Cubic on FA	-0.024	0.372	0.294	0.277	0.213
	(0.020)	(0.024)	(0.031)	(0.039)	(0.053)
Binary measure, 50% treated	-0.004	0.111	0.107	0.115	0.078
	(0.004)	(0.008)	(0.009)	(0.010)	(0.013)

Brazil estimates: effective minimum wage

	Emp.	Outcome		
		p10 - p50	p25 - p50	p90 - p50
Structural model	-0.033	0.208	0.070	0.005
Effective min. wage	-0.061 (0.022)	0.188 (0.020)	0.127 (0.015)	0.076 (0.033)
Effective min. wage, no region FE	-0.090 (0.009)	0.158 (0.014)	0.091 (0.009)	-0.059 (0.022)
Effective min. wage, no time FE	0.101 (0.004)	0.200 (0.006)	0.103 (0.005)	-0.127 (0.013)
AMS, no time FE	0.118 (0.006)	0.201 (0.006)	0.101 (0.005)	-0.149 (0.013)